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Galactic panel

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Keeping an eye on the brightest dusty white dwarf: Ground-based support for JWST observations

Abstract

Polluted white dwarfs with dusty disks are essential laboratories for studying the evolution of planetary systems and the ultimate fate of exoplanets. The pulsating white dwarf G29-38 is the prototype of such systems and remains the only one that has been extensively characterized across multiple wavelengths. We propose a long-term monitoring campaign to perform a detailed analysis of its pulsations. These observations will enable the first reverberation mapping of a debris disk around a white dwarf, offering unprecedented insight into the disk's structure and its response to stellar variability-an effort that will be crucial for the mineralogical modeling required by ongoing JWST observations. The SPECULOOS telescope is ideally suited for this program, providing the high-cadence photometry needed to capture even the shortest-period pulsation modes.

Observing Blocks

Instrument/Telescope	Req. time	Min. time	1 st Option	2 nd Option
CCD Speculoos/Speculoos	18 nights	15 nights	November Any	October Any

Cols

Name	Institution	e-mail	Observer?
Tim Cunningham	OnCL	tim.cunningham@cfa.harvard.edu	False
Joaquin Meza	USM	joaquin.mezal@usm.cl	False

Status of the project

- Past nights: 0
- Future nights: 36
- Long term: True

- Large program: False
- Thesis: False

List of Targets

ID	RA	DEC	Mag
G29-38	23:28:47.6365087803	+05:14:54.235051503	V=13.00

SCIENTIFIC AIM AND RATIONALE

Astrophysical context: It is well established that 25–50% of white dwarfs (WDs) exhibit metal pollution in their atmospheres [15], and it is now widely accepted that the primary source of this pollution is the remnants of evolved exoplanetesimals—such as asteroids, comets, and planets [7, 5]. This high incidence is consistent with the observational estimate that, on average, each main-sequence star hosts at least one exoplanet [19]. Furthermore, a significant fraction (10–20%) of metal-polluted WDs also exhibit infrared excesses, attributed to dusty disks formed from the tidal disruption of planetary bodies [29, 17, 2, 8, 4]. Due to the high surface gravity of WDs, heavy elements sink rapidly out of their typically pure H or He atmospheres via microscopic diffusion [22]. To interpret observed metal abundances, the accretion–diffusion framework is employed, where the removal of heavy elements occurs over timescales ranging from days to millions of years, depending on the WD’s properties [14, 10]. By combining a precise WD atmospheric model [16] with measured spectroscopic metal abundances, researchers can infer both the accretion rate and the composition of the disrupted parent body. Metal abundances, accretion rates, and disk lifetimes calculations for hundreds of polluted WDs rely on atmospheric analyses of optical [30, 12] and UV [15, 7, 18] spectra. With hundreds of HST orbits [7, 15] and dozens of VLT/Keck nights [30, 12] already devoted to this effort, the study of metal-polluted WDs has become a mature and active field—now encompassing the composition analysis of over a thousand disrupted planetary bodies.

Why G29-38? Out of 1700 metal-enriched WDs [26], the prototypical polluted WD G29-38 had the first discovered debris disk [9, 11] and it is a prime target for an extensive JWST study for the following reasons (Figure 1): **i)** It is the closest ($D = 17.54 \pm 0.02$ pc) and brightest ($W_2 = 10.7$ mag) WD currently accreting from a debris disk, **ii)** It is one of the best studied WDs having been targeted extensively with optical [27] and UV [14] spectroscopy, including a recent 35 orbits of HST/COS time (ID 15839), meaning that the photospheric metal abundances and WD parameters are exceptionally well constrained, **iii)** It has the best-modelled debris disk thanks to extensive multi-wavelength data [20, 6, 28, 1, 31], and **iv)** It is the *only* WD with an instantaneous accretion rate measured from X-ray observations [3], and **v)** It is currently being targeted as part of a 5-Cycle, 615 ks X-ray monitoring program with the Chandra X-ray Observatory, which provides the first opportunity to study accretion rate variability in a polluted WD, and **vi)** It is being targeted with 11.9-hr JWST Cycle 4 program (9077) to provide unprecedented constraints on the composition and variability of the debris from 1–24 μm , and it is this final point the motivation for a large optical ground-based monitoring campaign.

A JWST program: The Cycle 4 JWST observations G29-38 include spectroscopy with MIRI/MRS high-cadence low resolution time-series spectroscopy with NIRSpec and MIRI/LRS. The medium-resolution spectroscopy will yield the cornerstone spectrum which will transform the interpretation of the vast sample of polluted WDs already observed with JWST (>600 hours now awarded to evolved planetary system science) through the characterization of new mineralogical features detected at high S/N. The time-series data will allow us to investigate IR variability already detected in the disk. Our observations will allow us to separate variability from the reflection or reprocessing of the WD pulsations, and search for intrinsic IR variability in the disk from for instance the collisional destruction of dust grains. This will enable the first reverberation mapping of the disk, exploiting the light travel time to reconstruct the first geometric model of a WD debris disk. This ambitious program requires simultaneous high-precision optical lightcurves to constrain a detailed asteroseismological modeling of the stochastic WD surface temperature fluctuations. This model will be indispensable in the interpretation of our rich JWST dataset, thus SPECULOOS ground-based program is crucial.

Work plan with SPECULOOS: Variability G29-38 is variable in the optical because of stellar (non-radial) pulsations [13, Fig. 2], and we must measure that variability in the JWST regime to calibrate the WD’s contribution and get a clean disk spectrum. In addition, there may be stochastic short-term variability, due for example to collisions in the disk [23, 28], which could have not

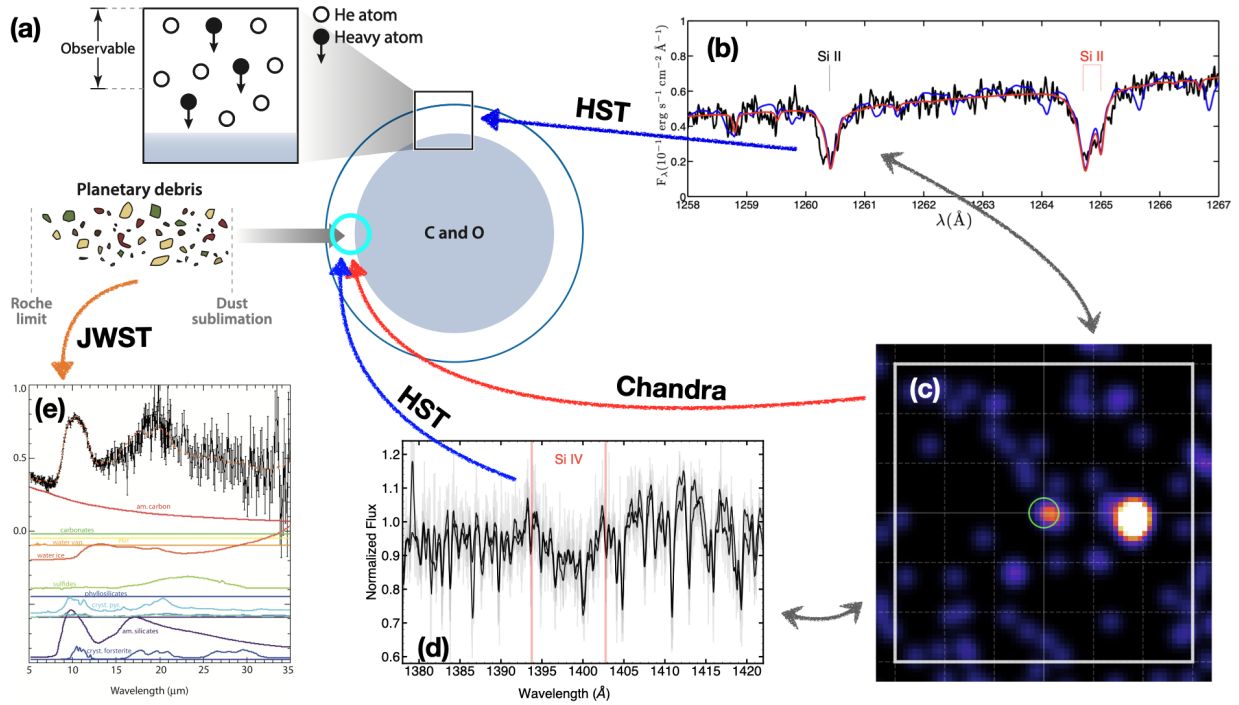


Figure 1: Overview of the ongoing multi-wavelength observations being leveraged to probe the physics of accretion onto our target, G29-38: the prototype for all metal-polluted WDs. *Clockwise from top-left*: (a): In the photosphere of a WD, heavy elements sink on timescales of few days to years. The metals persistently detected in some WDs result from material accreting from a debris disk formed by the disruption of planetary remnants. (b): The average COS spectrum of G29-38 (black) fitted with an atmosphere model of an effective temperature of 11820 K (red) clearly shows absorption features of Si II [27]. The COS spectrum also contains absorption features associated with C I & II and O I; (c): *Chandra* ACIS observation [3]. Recorded counts in soft+medium band (0.5–2.0 keV) convolved with instrumental PSF. *Chandra* HRC observations are approved for Cycles 24–27, providing X-ray observations of our target over a more than twenty year baseline; (d): a recently identified single-peaked high-ionization emission feature on the Si IV doublet with archival COS observations. This feature likely originates from the X-ray-emitting region, as the low-energy tail of the accretion-induced emission; (e): *Spitzer* spectroscopy of the debris disk around our target [21]. Mineralogical modelling allows to infer the composition of the disk, which will be highly improved with the JWST data.

been probed in detail prior to JWST. The optical pulsations of the WD have periods in the range 100–1240 s [13] and amplitudes up to 30% (with a mean amplitude of 5%). Recently, NIR periodic modulations have been identified below $2.5 \mu\text{m}$ [31] with simultaneous NIR spectroscopy with SpeX and optical photometry. Below $2 \mu\text{m}$, periods were consistent with reprocessing of the WD pulsation periods, whilst between 2 and $2.5 \mu\text{m}$, the periods diverged from the dominant pulsation modes; evidence of intrinsic variability within the disk. To understand the nature of this IR variability, it is essential to go to longer wavelengths, inaccessible from the ground. Our target is sufficiently bright in the IR that time-series NIRSpec/PRISM and MIRI/LRS spectroscopy offers a unique opportunity to detect variability on short timescales at longer wavelengths and test its origin. We will also lead an extensive ground-based optical photometric survey to simultaneously measure the WD optical flux. The system is near the celestial equator (Dec=+05 deg), and it is thus well placed for observers both in hemispheres. In addition to SPECULOOS, we will also arrange for continuous optical photometric time-series via an established network of observers owning small aperture (30–40 cm) telescopes, sufficient to obtain high-quality time-resolved light curves of the pulsations for such a bright target ($V = 13 \text{ mag}$). G29-38 is scheduled in TESS Sector 92 in May 2025. Unfortunately this does not overlap with the visibility of our target by JWST and thus an intensive ground-based program is essential. Nonetheless, the proposed observations will be combined with the TESS Sector 92 data to

establish a baseline of the pulsation mode stability over a timescale of months, depending on when our target is scheduled by JWST.

We propose to obtain one month high-cadence optical photometry with SPECULOOS, simultaneous with Cycle 4 JWST observations of the iconic debris disk around G29–38. The proposed program will 1) provide essential calibration for the JWST observations, allowing us to accurately model the WD pulsations, and 2) significantly enhance the science output, allowing to attempt the first reverberation mapping of a WD debris disk. Through this synergistic program, we will measure with high precision the disk composition and the location of the coldest emitting dust. This will address three key open questions: 1) how much material in the disk has undergone severe thermal alterations? 2) to what radius does the disk extend? and 3) is the disk variable at MIR wavelengths?

Imaging capabilities of the 1m telescopes are perfectly suited to gain high-precision optical lightcurves of G29-38. The pulsations have amplitudes of 0.1 magnitudes in the optical and thus we will easily reach the flux precision required to reach our science goals. The success of this project relies on the identification of the l and m modes that are active during the JWST observations, which will enable us to reconstruct the exact location of the temperature variation on surface. This information can be obtained from the proposed ground-based photometry: Tucker et al. [24] accurately predicted the shape of the ultraviolet pulsations of WD 1401-147 observed with *GALEX* by modeling the quasi-simultaneous optical Whole Earth Telescope light curve. Recently, with time-series NIR spectroscopic observations with SpeX, [31] detected variability at all wavelengths between 0.7–2.5 μm . The demonstrated that simultaneous observations with the Whole Earth Telescope were adequate to confirm that at shorter wavelengths ($<2\mu\text{m}$), the variability is consistent with the WD pulsations, but that at longer wavelengths the variability corresponds to variability in the debris disk. This demonstrates the promise of our JWST program (which will cover 1–15 μm with time-series spectroscopy), coupled with continuous 1m-class ground-based photometric monitoring, to unveil new insights about the nature and origin of intrinsic variability in WD debris disks.

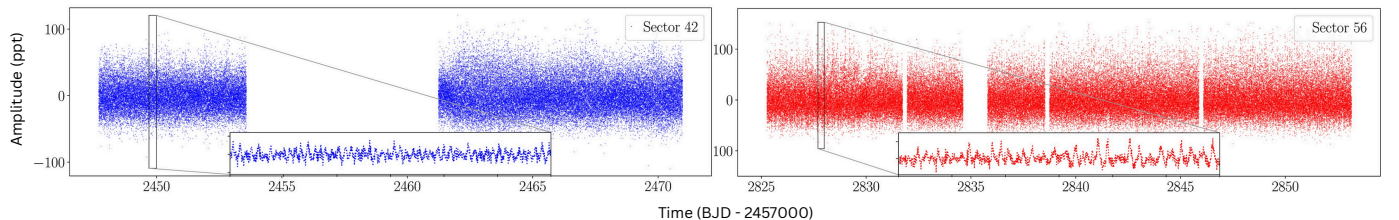


Figure 2: *TESS* Light curves revealing the variability caused by the non-radial pulsations of the white dwarf G29-38 require integration times shorter than 20 seconds in order to properly sample and resolve each pulsation mode, which is possible with SPECULOOS. Adapted from [25]

References

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Observing Strategy:

The white dwarf G29-38 exhibits pulsation periods ranging from approximately 120 to 1,000 seconds [e.g. 13]. Its brightness makes it an ideal target for high-precision photometry aimed at resolving its variability in detail.

To calculate the exposure time, we incorporated a white dwarf spectrum [$\log g = 8.0$ and $T_{\text{eff}}=12,000$ K; 25] to the SPOCK_CHILEAN tool. With an exposure time of 10 seconds, we expect to reach up to 38,000 ADU in the g' filter, which covers the H_{β} line. The selection of the g' filter, is given that it is the bluest filter available, and the pulsation amplitudes are expected to be slightly higher towards the bluer wavelengths, allowing for an excellent SNR in the resulting light curves.

The combined readout and overhead time is approximately 10 seconds, so we will obtain one data point every 20 seconds. This cadence ensures adequate temporal sampling of even the shortest pulsation periods (e.g. 120 seconds). Standard calibration procedures will be applied, and binning will not be used. Due to the brightness of the target, lunar phase constraints are not a concern.

These high-quality light curves will enable us to resolve the pulsation periods and study their amplitudes. This is essential for constructing accurate pulsation models, which will, in turn, help us identify short-term variations in the system and determine whether they originate from pulsations or other phenomena such as dust disk collisions.

In summary, **we request 18 CNTAC nights for SPECULOOS photometric observations during October and November. The November observations are particularly critical to enable simultaneous coverage with JWST** (see Long Term Status), coinciding with the period when G29-38 is optimally visible during the semester.

The images to produced the differential lightcurve will be reduced with the python code PROSE.

Table 1: Information of the target

target name	RA (deg)	DEC (deg)	g (mag)
G29-38	352.1984854532512	+05.2483986254175	13.054 ± 0.007

The Phase II for JWST has not yet be finalized, and thus the precise timing of the observations is not known. For now, the observations may be scheduled in three windows: 1) 10-Jun-to 02-Aug 2025, 2) 02-Nov to 21-Dec 2025 and 3) 11-Jun to 02-Aug 2026. To increase ground-based visibility, have requested to reduce these windows to late July and early November in 2025, or later July in 2026. **We require Time Critical because we want a good chance of getting scheduled during the JWST observations.** We also submit the request for the following 3 semesters; 2025B, 2026A and 2026B to ensure that we have the necessary coverage in each of the possible schedulable windows. The absolute priority is the one week of observations in which the middle is fully simultaneous with our JWST observations. However, just as with the TESS data, the observations in semesters before or after the JWST observations will still be scientifically valuable for the characterization of the pulsation mode stability over the timescale of months to one year.

The pulsations span periods ranging from 120 to 1000 seconds, with 56 distinct modes identified in recent TESS observations [25, see Fig.,2]. These modes interact with one another, producing a highly stochastic and complex light curve. However, not all periodicities are excited simultaneously, resulting in continuous variations in the star's pulsational behavior [13]. Our program will in the first instance provide the unique time-dependent finger print of these pulsations from the WD. This will enable us to search in the high-S/N JWST time-series spectroscopy for wavelength-dependent reflection. Since longer wavelength emission comes for further reaches of the debris disk, the wavelength dependence of the reflection will allow us to infer a light travel time to different radial positions in the disk, enabling the first reverberation mapping of a WD debris disk. Beyond the strictly simultaneous optical lightcurves, we also require a long continuous baseline either side of the JWST observations. Due to the complex nature of the pulsations, a long baseline is essential for the development of a robust asteroseismological model of the white dwarf surface temperature. This model will yield a time-dependent WD effective temperature which will be essential for constraining the slope of the WD SED in the NIR. This will be critical for removing the white dwarf contribution from the JWST/NIRSpec spectra, allowing us to search for intrinsic variability from the debris disk, either from reflection or reprocessing of radiation from the pulsating WD, or from short-term dust and gas production in the disk from grain collisions. It is for this reason that high-cadence time-series JWST spectroscopy has been awarded. The richness of the upcoming JWST dataset cannot be understated. It is essential that we establish a robust program of continuous, high-cadence optical imaging, and SPECULOOS is the perfect facility.

JWST *provides the only means to study dust mineralogy.* Dust mineralogy in white dwarf debris disks provides insights into whether the material has undergone significant thermal processing. Recent JWST observations have detected features such as silica dust, crystalline silicates, and possible carbonates, suggesting both current disk activity and inherited material from planetesimals. Notably, JWST/MIRI spectroscopy of systems like WD 0145+234 reveals silicate features consistent with steady-state collisional evolution and hints of aqueous alteration [1]. In G29–38, previously undetectable PAH features may now be within reach thanks to JWST’s superior sensitivity. Shocks may transform amorphous grains into crystalline forms, altering the emission features observed around $10\mu\text{m}$. To distinguish between crystalline and amorphous grain contributions and measure dust abundances more precisely, high-resolution spectroscopy is essential.

There are several observational programs targeting G29-38 in addition to the JWST observations.

Notably, a multi-cycle 615 ks X-ray monitoring program with the **Chandra X-ray Observatory** (Program IDs: 22200039, 24200446, 25200153) is underway. This program aims to track variability in the accretion rate of this polluted white dwarf. X-ray data are particularly valuable in this context, as they provide a direct, model-independent measurement of the accretion rate — unlike other observational methods that rely more heavily on modelling.

A 35-orbit **HST/COS** program (ID 15839) has been conducted, providing exceptionally well-constrained measurements of both the photospheric metal abundances and the white dwarf parameters (analysis in preparation). The ongoing data analysis has revealed Si IV emission (see panel d in Fig. 1), which is highly unusual and has not been observed in any other white dwarf, except in systems exhibiting double-peaked profiles associated with gaseous discs [17]. However, our plasma modelling of the thermal structure suggests that this emission may originate from a region very close to the white dwarf photosphere, rather than from a surrounding gas disc.

Chilean involvement This proposal is led by a team of three researchers:

- O. Toloza (PI of this proposal, UTFSM) also serves as the Principal Investigator of the HST program focused on UV spectroscopic modeling [e.g. 2]. In this project, she will lead the reduction and analysis of optical light curves obtained with SPECULOOS and perform Lomb-Scargle periodogram analysis to identify periodic signals.

- J. Meza, a second-year PhD student at UTFSM. He will be preparing the observing blocks (OBs), drawing on prior experience with his previous SPECULOOS program. Although these data are not directly part of his thesis, this contribution reflects his involvement in a broader collaborative effort.

- T. Cunningham (Cfa Harvard), he discovered G29-38 as a new class of X-ray emitters [3]. He is the PI of the JWST and the Chandra programs. He will carry out the mineralogical analysis of the MIRI spectroscopy based on the elemental abundances derived from the UV spectroscopy, and the accretion rate variability study from the Chandra data [3].

Relevant publications

- [P1] A. Swan, J. Farihi, K. Y. L. Su, and S. J. Desch “The first white dwarf debris disc observed by JWST”, [MNRAS](#), vol. 529, Mar. 2024.
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- [P3] T. Cunningham, P. J. Wheatley, P.-E. Tremblay, B. T. Gänsicke, G. W. King, O. Toloza, and D. Veras “A white dwarf accreting planetary material determined from X-ray observations”, [Nature](#), vol. 602, Feb. 2022.